Objectives

This project aims to design and implement an enterprise-grade network topology fulfilling the following objectives:

- Design a network with 2 core routers (R1, R2) using EIGRP dynamic routing protocol for automatic path calculation and convergence.
- Configure static backup routes with higher administrative distance (AD 150) to provide failover redundancy when EIGRP paths fail.
- Deploy 4 VLANs (VLAN 10, 20, 30, 40) across multiple access layer switches to achieve proper traffic segmentation.
- Set up Inter-VLAN routing using the Router-on-a-Stick method with 802.1Q trucking on sub interfaces.
- Configure DHCP relay agents (ip helper-address) so centralized DHCP server provides IPs to all VLANs.
- Add and configure DNS and Web servers in a dedicated server segment.
- Test and verify both primary path (EIGRP) and backup path (static routes) selection during link failure scenarios.

1. Theory

1.1 EIGRP (Enhanced Interior Gateway Routing Protocol)

EIGRP is an advanced distance-vector routing protocol developed by Cisco. It uses the Diffusing Update Algorithm (DUAL) to compute loop-free paths and rapid convergence. EIGRP uses bandwidth and delay as its composite metric by default, and routers only send updates when topology changes occur (partial updates), making it more efficient than protocols like RIP. EIGRP has a default administrative distance of 90 for internal routes, and it supports VLSM, classless routing, and route summarization.

1.2 Static Routes and Administrative Distance

Static routes are manually configured paths in the routing table. When used alongside dynamic protocols like EIGRP, they can serve as backup routes by assigning a higher administrative distance. Since EIGRP routes have an AD of 90, static backup routes are configured with AD 150. The router installs only the lowest AD route in the routing table; the static route becomes active only when the EIGRP-learned route disappears (i.e., a link goes down), providing automatic failover.

1.3 VLANs and 802.1Q Trunking

A VLAN (Virtual Local Area Network) logically segments a switch into multiple broadcast domains. Traffic from different VLANs is isolated, improving security and network performance. 802.1Q is the IEEE standard for VLAN tagging on trunk links; it adds a 4-byte tag to Ethernet frames so multiple VLANs can traverse a single physical link between a switch and a router.

1.4 Inter-VLAN Routing (Router-on-a-Stick)

Devices in different VLANs require a Layer 3 device to communicate. Router-on-a-Stick uses a single physical router interface connected to a trunk port on a switch. Logical subinterfaces (e.g., Gi0/2.10, Gi0/2.20) are created, each configured with dot1Q encapsulation for the corresponding VLAN ID and assigned an IP address that acts as the default gateway for that VLAN.

1.5 DHCP Relay Agent

DHCP operates via broadcast, which cannot cross router boundaries. A DHCP relay agent (configured with the ip helper-address command) converts DHCP broadcast messages from clients into unicast packets directed at the centralized DHCP server. This allows a single DHCP server to serve multiple VLANs without placing a server in each segment.

1.6 DNS and Web Servers

A DNS (Domain Name System) server translates human-readable hostnames (e.g., www.mynetwork.com) into IP addresses. A Web server hosts HTTP/HTTPS services accessible by clients. Both are placed in a dedicated server segment for centralized management and reachability from all VLANs through the routed network.

2. Network Topology Overview

The network employs a triangular router topology (R1, R2, R3) for core redundancy, with R2 and R3 connecting to access switches (SW1, SW2) that host end-user devices. A dedicated server switch (SW-Server) connected to R1 hosts critical network services. This design ensures that if a primary link or router fails, an alternative path is available, maintaining network connectivity.

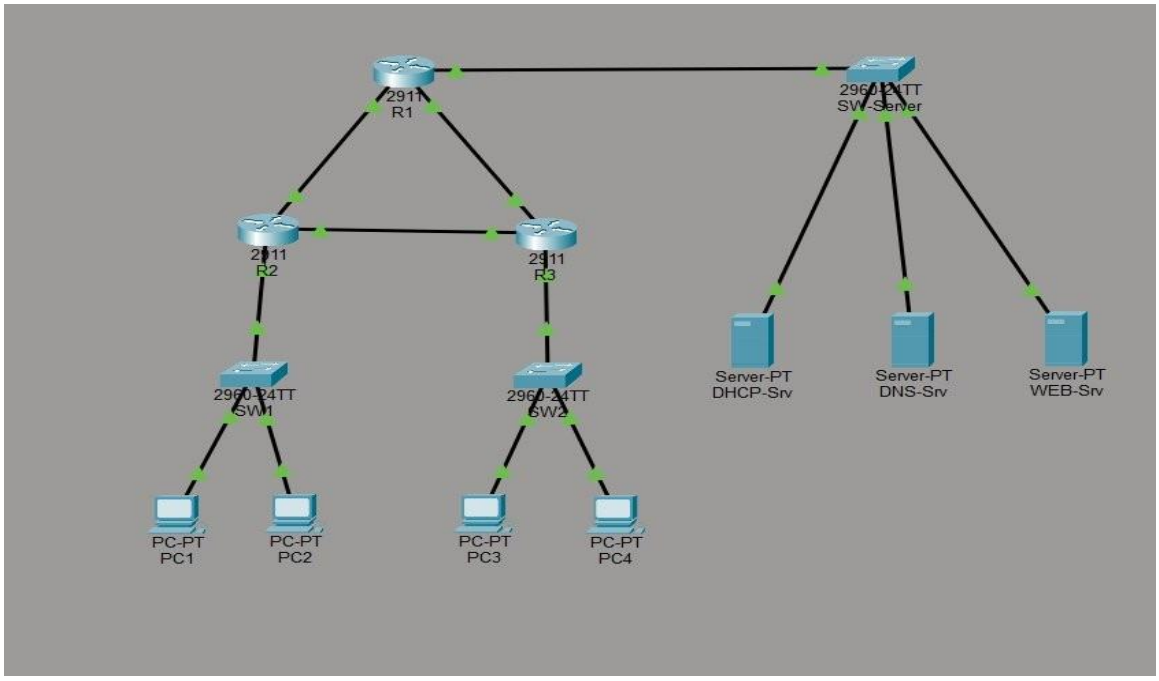

Figure 1: Network Topology Diagram

2.1 Architecture Highlights

Component	Function	Redundancy
Core Routers (R1, R2, R3)	Full mesh routing with EIGRP	Triple path redundancy
Access Switches (SW1, SW2)	VLAN segmentation and access	Dual-homed to routers
Server Switch	Centralized services hosting	Connected to core router R1
DHCP/DNS/Web Servers	Network services	Accessible via all paths

3. IP Addressing Scheme

The network utilizes private IP address space with careful subnet allocation to support current requirements and future growth. The addressing scheme follows industry best practices for hierarchical IP design.

3.1 VLAN Subnets

VLAN ID	VLAN Name	Network	Subnet Mask	Default Gateway
10	PC1_VLAN	192.168.10.0	255.255.255.0	192.168.10.1 (R2)
20	PC2_VLAN	192.168.20.0	255.255.255.0	192.168.20.1 (R2)
30	PC3_VLAN	192.168.30.0	255.255.255.0	192.168.30.1 (R3)
40	PC4_VLAN	192.168.40.0	255.255.255.0	192.168.40.1 (R3)

3.2 Inter-Router Links

Point-to-point links between routers utilize /30 subnets (4 addresses) for efficient IP address utilization. This follows Cisco best practices for router interconnections.

Link	Interface	IP Address	Network
R1-R2	R1: Gig0/0 R2: Gig0/0	10.1.12.1 10.1.12.2	10.1.12.0/30
R1-R3	R1: Gig0/1 R3: Gig0/0	10.1.13.1 10.1.13.2	10.1.13.0/30
R2-R3	R2: Gig0/1 R3: Gig0/1	10.1.23.1 10.1.23.2	10.1.23.0/30

3.3 Server Segment

Device	IP Address	Subnet Mask	Default Gateway	Purpose
R1 (Gig0/2)	192.168.100.1	255.255.255.0	-	Server gateway
DHCP Server	192.168.100.10	255.255.255.0	192.168.100.1	IP address management
DNS Server	192.168.100.11	255.255.255.0	192.168.100.1	Name resolution
Web Server	192.168.100.12	255.255.255.0	192.168.100.1	HTTP/HTTPS services

4. Router Configurations

All routers are configured with EIGRP (Autonomous System 100) for dynamic routing and static backup routes with Administrative Distance of 150 to ensure EIGRP-learned routes are preferred.

4.1 Router R1 - Core Router

R1 serves as the core router connecting to both edge routers (R2, R3) and the server segment. It advertises server networks and maintains backup routes to all VLANs.

```
hostname R1 ! interface
GigabitEthernet0/0  description Link
to R2  ip address 10.1.12.1
255.255.255.252 !
interface GigabitEthernet0/1
description Link to R3  ip address
10.1.13.1 255.255.255.252 !
interface GigabitEthernet0/2  description Server
Segment  ip address 192.168.100.1 255.255.255.0
! router eigrp 100  network 10.1.12.0 0.0.0.3
network 10.1.13.0 0.0.0.3  network 192.168.100.0
0.0.0.255  no auto-summary ! ! Static backup
routes with AD 150 ip route 192.168.10.0
255.255.255.0 10.1.12.2 150 ip route
192.168.20.0 255.255.255.0 10.1.12.2 150 ip
route 192.168.30.0 255.255.255.0 10.1.13.2 150
ip route 192.168.40.0 255.255.255.0 10.1.13.2
150
```

4.2 Router R2 - Edge Router with VLANs 10 & 20

R2 provides inter-VLAN routing for VLANs 10 and 20 using router-on-a-stick configuration. DHCP relay agents forward client requests to the centralized DHCP server.

```
hostname R2 ! interface
GigabitEthernet0/0
description Link to R1
 ip address 10.1.12.2 255.255.255.252
!
interface GigabitEthernet0/1
description Link to R3  ip address
10.1.23.1 255.255.255.252 !
interface GigabitEthernet0/2
description Trunk to SW1  no ip
address !
interface GigabitEthernet0/2.10
description VLAN 10 - PC1
encapsulation dot1Q 10  ip address
192.168.10.1 255.255.255.0  ip
helper-address 192.168.100.10 !
interface GigabitEthernet0/2.20
description VLAN 20 - PC2
encapsulation dot1Q 20  ip address
192.168.20.1 255.255.255.0  ip
helper-address 192.168.100.10 !
router eigrp 100  network 10.1.12.0
0.0.0.3  network 10.1.23.0 0.0.0.3
network 192.168.10.0 0.0.0.255
network 192.168.20.0 0.0.0.255  no
auto-summary ! ! Backup routes to
R3's VLANs ip route 192.168.30.0
255.255.255.0 10.1.23.2 150 ip route
192.168.40.0 255.255.255.0 10.1.23.2
150
```

4.3 Router R3 - Edge Router with VLANs 30 & 40

R3 provides inter-VLAN routing for VLANs 30 and 40 with similar configuration to R2, ensuring consistent network design and ease of management.

```
hostname R3 ! interface
GigabitEthernet0/0  description Link
to R1  ip address 10.1.13.2
255.255.255.252 ! interface
GigabitEthernet0/1  description Link
to R2  ip address 10.1.23.2
255.255.255.252 ! interface
GigabitEthernet0/2  description
Trunk to SW2  no ip address !
interface GigabitEthernet0/2.30  description
VLAN 30 - PC3  encapsulation dot1Q 30  ip
address 192.168.30.1 255.255.255.0  ip helper-
address 192.168.100.10 ! interface
GigabitEthernet0/2.40  description VLAN 40 - PC4
encapsulation dot1Q 40  ip address 192.168.40.1
255.255.255.0  ip helper-address 192.168.100.10
! router eigrp 100  network 10.1.13.0 0.0.0.3
network 10.1.23.0 0.0.0.3  network 192.168.30.0
0.0.0.255  network 192.168.40.0 0.0.0.255  no
auto-summary ! ! Backup routes to R2's VLANs ip
route 192.168.10.0 255.255.255.0 10.1.23.1 150
ip route 192.168.20.0 255.255.255.0 10.1.23.1
150
```

5. Switch Configurations

Access layer switches provide VLAN segmentation and connect end-user devices to the network. Trunk ports use 802.1Q encapsulation to carry multiple VLANs to routers.

5.1 Switch SW1 - Access Layer for VLANs 10 & 20

```
hostname SW1
!
vlan 10  name
PC1_VLAN !
vlan 20  name
PC2_VLAN !
interface FastEthernet0/1
description Trunk to R2
switchport mode trunk !
interface FastEthernet0/2
description PC1 Access Port
switchport mode access
switchport access vlan 10
spanning-tree portfast !
interface FastEthernet0/3
description PC2 Access Port
switchport mode access
switchport access vlan 20
spanning-tree portfast
```

5.2 Switch SW2 - Access Layer for VLANs 30 & 40

```
hostname SW2
vlan 30  name
PC3_VLAN !
vlan 40  name
PC4_VLAN !
interface FastEthernet0/1
description Trunk to R3
switchport mode trunk !
interface FastEthernet0/2
description PC3 Access Port
switchport mode access
switchport access vlan 30
spanning-tree portfast !
interface FastEthernet0/3
description PC4 Access Port
switchport mode access
switchport access vlan 40
spanning-tree portfast
```

5.3 Switch SW-Server - Server Segment Access

The server switch provides dedicated connectivity for critical network services. All servers reside in a single VLAN for simplified management.

```
hostname SW-Server
!
vlan 100  name
Server_VLAN !
interface FastEthernet0/1
description Uplink to R1
switchport mode access
switchport access vlan 100
!
interface FastEthernet0/2
description DHCP Server
switchport mode access
switchport access vlan 100
!
interface FastEthernet0/3
description DNS Server
switchport mode access
switchport access vlan 100
!
interface FastEthernet0/4
description Web Server
switchport mode access
switchport access vlan 100
```

6. Server Configurations

6.1 DHCP Server Configuration

The DHCP server is configured with four address pools, one for each VLAN. Each pool provides IP addresses, default gateway information, and DNS server addresses to clients.

Pool Name	Network	Start IP	Default Gateway	DNS Server
VLAN10	192.168.10.0/24	192.168.10.50	192.168.10.1	192.168.100.11
VLAN20	192.168.20.0/24	192.168.20.50	192.168.20.1	192.168.100.11
VLAN30	192.168.30.0/24	192.168.30.50	192.168.30.1	192.168.100.11
VLAN40	192.168.40.0/24	192.168.40.50	192.168.40.1	192.168.100.11

6.2 DNS Server Configuration

DNS service is enabled with A record configured for web server name resolution. Additional records can be added as network services expand.

Record Type	Hostname	IP Address
A Record	www.mynetwork.com	192.168.100.12

6.3 Web Server Configuration

- HTTP Service: Enabled
- HTTPS Service: Enabled
- Custom index.html page deployed with network welcome message
- Accessible via domain name: www.mynetwork.com

7. Verification and Testing

Comprehensive testing validates all network functionality including connectivity, services, and redundancy mechanisms.

7.1 End-to-End Connectivity Tests

DHCP Functionality:

- All PCs successfully obtain IP addresses from DHCP server
- Correct default gateway and DNS server assigned to each VLAN

• IP addresses allocated from configured ranges (starting at .50)

Inter-VLAN Communication:

• PCs can ping their default gateway

• PCs in different VLANs can communicate (e.g., PC1 → PC3)

• All PCs can reach server segment (DHCP, DNS, Web servers)

7.2 Service Validation

DNS Resolution Test:

• From any PC, ping www.mynetwork.com resolves to 192.168.100.12

• DNS queries properly forwarded and resolved

Web Server Access:

• Web browser successfully loads www.mynetwork.com

• Custom welcome page displays correctly

• Both HTTP and HTTPS services functional

7.3 Redundancy and Failover Testing

The following procedure validates primary and backup path selection:

Step	Action	Expected Result
1	From PC1, start continuous ping to PC3 (ping 192.168.30.10 -t)	Successful ping responses Path: PC1 → R2 → R1 → R3 → PC3
2	Shut down R1-R2 link (R1: int g0/0, shutdown)	2-3 dropped packets during EIGRP reconvergence Traffic reroutes via R2 → R3 → R1
3	Observe routing table on R2 (show ip route)	Static route to PC3 VLAN activates Route via 10.1.23.2 (R3) appears
4	Re-enable R1-R2 link (R1: int g0/0, no shutdown)	Brief interruption EIGRP route re-established Traffic returns to primary path

Key Observations:

• EIGRP convergence time: approximately 1-3 seconds

• Static routes remain in standby with AD 150 (EIGRP default AD is 90)

• Network maintains connectivity during single link failures

• Automatic failback to primary path when link restored

8. Conclusion

This network implementation successfully fulfills all project objectives and demonstrates enterprise-grade capabilities for a multi-segment, redundant network environment using two routers.

The deployment showcases:

- Redundant routing — EIGRP dynamic routing with static backup routes (AD 150) ensures automatic failover with minimal packet loss during link failures.
- Network segmentation — Four VLANs properly segment user traffic, with 802.1Q trunking efficiently carrying multiple VLANs over single physical links.
- Centralized services — DHCP relay agents enable IP address management for all VLANs from a single server. DNS provides hostname resolution and the web server delivers HTTP/HTTPS services from all segments.
- Verified redundancy — Testing confirmed automatic failover and recovery, with AD-based route selection ensuring optimal path usage at all times.

The architecture provides a solid foundation for growth, with capacity for additional VLANs, services, and further redundancy enhancements.